

- 2 (a) (i) e.g. (phase) change from liquid to gas / vapour
thermal energy required to maintain constant temperature B1 [1]
(do not allow 'convert water to steam')
- (ii) e.g. evaporation takes place at surface B1
boiling takes place in body of the liquid B1
e.g. evaporation occurs at all temperatures B1
boiling occurs at one temperature B1 [4]
- (b) (i) volume = $(\frac{48}{4.5})$ 10.7 cm³ A1 [1]
- (ii) 1 volume = $10.7 / (6.0 \times 10^{23})$
= 1.8×10^{-23} cm³ A1 [1]
2 separation = $\sqrt[3]{(1.8 \times 10^{-23})}$
= 2.6×10^{-8} cm A1 [1]

[Total: 8]

- 3 (a) (i) speed = 4.0 m s^{-1} (*allow 1 s f*) A1 [1]